

Talk to You More, Listen to You More, and Feel Better in Your Mind: Utilizing Self-talk as a Mechanism of Action for Alleviating Anxiety

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Abstract

While self-talk represents a promising Mechanism of Action (MoA) for anxiety management, the unfamiliarity of hearing one's own voice poses significant challenges for therapeutic engagement. Through two empirical studies ($N=60$, $N=119$), we examined a novel digital therapeutic approach combining Acceptance and Commitment Therapy (ACT) with self-talk externalization. Our first study confirmed self-talk's therapeutic potential but highlighted user discomfort with voice experiences. To address this, we developed a systematic voice equalization algorithm using Analytical Hierarchy Process (AHP) to identify optimal voice modifications through progressive refinement of 14 voice types. Our second study implementing this algorithm demonstrated both improved anxiety outcomes and enhanced user engagement. Participants reported that interacting with their modified voice facilitated deeper self-acceptance and emotional understanding. These findings suggest that self-talk, when properly integrated with voice equalization techniques, can effectively support anxiety reduction while promoting sustained therapeutic engagement in digital mental health interventions.

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CCS Concepts

• **Human-centered computing** → Human computer interaction (HCI); Empirical studies in HCI.

Keywords

Digital Therapeutics, Anxiety, Self-talk

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1 Introduction

Anxiety, which brings excessive worries and bodily tension, interferes with people's daily routines and capabilities [2, 14]. The World Health Organization notes that anxiety disorders affect 4% of the global population, which reveals their substantial impact [7, 10]. Without proper care, anxiety can disrupt one's responsibilities, social connections, and life satisfaction [38, 42, 46, 56]. Though various treatments exist - from essential techniques to medical approaches - the principal challenge lies in making these remedies available and suitable for more people.

The treatment of mild anxiety, which does not meet clinical thresholds, requires wider-reaching solutions yet lacks thorough research [31]. Digital Therapeutics (DTx) offer a viable path to deliver proven treatments to more people, which helps bridge the divide between care needs and access [16, 47, 49, 50]. ACT (Acceptance and Commitment Therapy) suits digital adaptation particularly well, as its organized framework and focus on mental adaptability work in its favor [41]. The key elements of ACT - mindful awareness, acceptance of thoughts, and actions that align with personal values - translate effectively to digital platforms, which allows for broad deployment without sacrificing treatment quality [1, 33, 53].

The success of DTx programs relies not only on choosing the right treatment approach but also on creating a user-friendly experience [32]. The field of Human-Computer Interaction (HCI) suggests that effective digital treatments must account for how users interact with and respond to the technology [55]. Our research examines self-talk as an innovative treatment method, which centers on using one's recorded voice as a therapeutic tool. Though self-talk typically occurs as inner dialogue, the act of recording it adds fresh therapeutic benefits. Yet many people feel uneasy when they hear their recorded voice, which might reduce their willingness to use such tools.

Our research tackles these obstacles by viewing self-talk in two ways: as a therapeutic tool (Mechanism of Action; MoA) and as a gentle push towards beneficial habits within digital treatments. We combined ACT principles with self-affirmation methods [11, 15, 34] to create a treatment that transforms self-talk into a helpful therapeutic tool. The treatment builds upon habits which people already have, rather than demanding entirely new ones. It places self-talk within ACT's structure, which may improve both user involvement and treatment results. We examined this new approach through two studies that explore:

- **RQ1.** To what extent our digital therapeutic approach, which employed self-talk techniques, demonstrates both effectiveness in reducing anxiety and feasibility in real-world application?
- **RQ2.** Which voice modification techniques prove most effective at boosting user involvement and therapeutic benefits within the ACT framework?

This systematic examination of these questions aims to enhance digital therapeutic approaches and expand our understanding of personalized digital mental health treatments.

2 Related Work

In this section, we explore psychological processes that enhance flexibility and shape therapeutic treatments, underpinning our research into self-talk as the main therapeutic tool (MoA) in digital treatment.

2.1 DTx for Strengthening Psychological Flexibility

Self-distancing, which lets people view emotions objectively, helps manage anxiety effectively [17, 20]. Within ACT, it serves as a core principle that helps people observe thoughts without being controlled by them [30, 30, 65]. Research shows specific language patterns, like using third-person pronouns, help regulate emotions

better [39]. Self-referencing involves processing information about oneself, activating key brain networks for emotional awareness. In treatment settings, it enhances emotional understanding and reduces reactivity to triggers [19, 63]. When combined with mindfulness, it improves emotional control [35, 59]. These mechanisms work together: self-distancing provides objective perspective while self-referencing enables deeper emotional processing [29, 48, 61]. Together, they help people stay aware of feelings without becoming overwhelmed, aligning well with ACT principles for anxiety management [20].

2.2 Self-talk as a Novel MoA of DTx

2.2.1 Self-talk and Anxiety. Hardy's (2006) work defines self-talk through five elements: self-directed statements, expression forms, content interpretation, temporal changes, and instructional/motivational functions [25]. Our research defines self-talk as the verbalization of one's thoughts and engaging with these verbalized expressions. Studies show voiced thoughts build resilience through improved self-esteem [9, 22, 23, 25, 44]. This resilience even offer protection against conditions like generalized anxiety disorder (GAD) by reducing anxiety escalation. Drawing on ACT principles [28, 60], we use recorded self-talk to strengthen self-observation, helping create psychological distance from unhelpful self-views [3, 64].

2.2.2 Self-talk as a Nudge in Behavioral Intervention for DTx. Nudging guides better decisions while preserving choice freedom [57]. In DTx, self-talk serves as an effective behavioral mechanism, helping reframe thoughts and emotional responses. Research shows it improves performance and reduces anxiety [28, 40, 54]. External self-talk through recording makes therapeutic processes concrete and more likely to trigger action [21]. In DTx design, self-talk serves dual purposes: encouraging treatment engagement and achieving therapeutic benefits, integrating behavioral science with HCI principles [5].

3 Method

We conducted two studies examining the efficacy of a self-talk-integrated DTx program for anxiety (Figure 1). Study 1 investigated the integration of self-talk with ACT and self-affirmation exercises, focusing on feasibility and acceptability. User feedback highlighted voice perception as a key challenge—people often feel uncomfortable hearing recordings of their own voice [37]. This insight led to the development of voice equalization to improve user experience. Building on these insights, our second study assessed (i) how well self-talk works as a treatment mechanism on its own and (ii) how it interacts with emotion-focused features to reduce anxiety. We used a 2×2 factorial design that compared four conditions: with and without self-talk, and with and without emotion-based features. This approach helped us understand how self-talk contributes to anxiety reduction and examine how voice adjustment affects user comfort, engagement, and treatment outcomes in a digital setting.

3.1 Study 1: Feasibility Test of Self-talk for DTx

3.1.1 Prototyping. We developed an iOS smartphone application using React Native framework. The system runs on a Node.js server

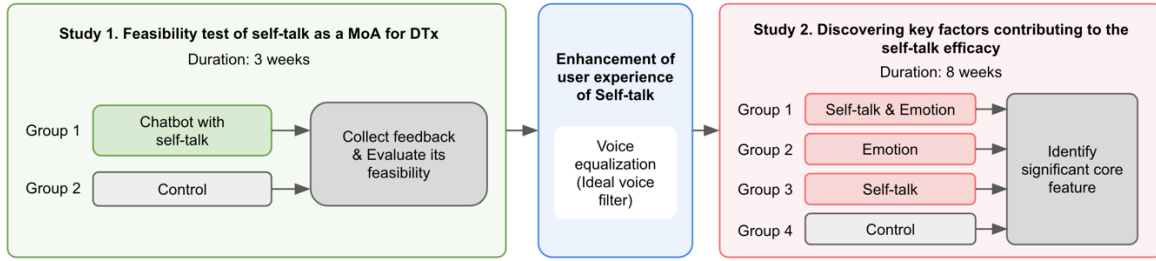


Figure 1: Stages of Research Development. Our research unfolded in two phases. Study 1 focused on developing a new intervention for anxiety reduction, followed by an exploratory study with two groups to test whether this novel approach would work in practice. Study 2 then addressed the user discomfort issues we identified and examined which factors make self-talk effective through a 2×2 study design (Self-talk vs. Non-self-talk × Emotion-based vs. Non-emotion-based).

deployed through Amazon EC2. Participants installed the application on their smartphones via TestFlight, Apple’s beta testing platform. The application collects contextual information about users to support self-distancing and self-referencing features, with particular focus on emotional states (Appendix A).

3.1.2 Procedure. To test whether self-talk works as a new therapeutic tool (i.e., MoA) for reducing anxiety in young adults, we conducted a three-week study with two groups. We recruited 60 participants (average age = 26.75 years, SD = 3.81; 18.3% male). After participants gave informed consent, we randomly assigned them to either the experimental or control group. The control group received an e-book application instead. Both groups used their applications once daily for 10 to 15 minutes per session. We designed the control group’s task to involve a smartphone – the same tool as the experimental group – but without the interactive self-talk features that our study examines. Participants received a gift card worth 35,000 KRW (about USD 30) after completing the study.

3.1.3 Data collection. We used the Beck Anxiety Inventory (BAI) to measure anxiety levels. The BAI, which is a self-report questionnaire that assesses anxiety severity [6], contains 21 items about anxiety-related thoughts, physical symptoms, and panic experiences.

3.2 Enhancement of Self-talk Experience: Voice equalization

Our preliminary study found that people often feel discomfort when hearing recordings of their voice. We therefore created a systematic comparison method that lets users find their preferred voice among several modified versions. The system uses the Analytical Hierarchy Process (AHP) algorithm to help users compare different voice modifications in pairs (Figure 2). AHP was chosen because it is a well-established multi-criteria decision-making (MCDM) method that effectively structures complex decision problems into a hierarchical model, allowing users to make systematic pairwise comparisons [67]. Compared to other MCDM approaches, AHP provides a more intuitive framework for subjective preference evaluation by quantifying users’ judgments and aggregating them into a final ranking [51, 67]. Through these comparisons, users can identify which voice modification sounds most natural to them. A study has shown that fundamental frequency (F0) and formant

structures are the two main factors that help people distinguish between their own voice and others’ voices [62]. Hartmann’s work further suggests that formant strongly influences how we perceive emotions in voices [26]. Based on these findings, we developed 14 different voice modifications for Study 2 by adjusting pitch and formant values (Appendix B).

3.3 Study 2: Discovering Key Factors Contributing to the Self-talk Efficacy

3.3.1 Prototyping. We developed a smartphone application that centers on emotion selection and self-talk features. We built the app for both Android and iOS platforms using Android Studio and Xcode, respectively. The application’s content reflects findings from our preliminary study. To improve the emotion-related features, we created simple illustrations of facial expressions. Research shows that emotions can be expressed through facial expressions, which fall into six main categories: neutral, happy, sad, fearful, angry, and surprised [18]. We excluded ‘surprised’ due to interpretation ambiguity and research suggesting the benefits of odd-numbered choices [4, 45]. These expressions allow categorization of user emotions into neutral, positive, or negative states.

3.3.2 Procedure. To determine which factors influence self-talk effectiveness, we conducted a study using a 2×2 design: Self-talk vs. Non-self-talk × Emotion-based vs. Non-emotion-based (Appendix C). The study ran for eight weeks, with the first week serving as an adaptation period to help users understand the process. We, afterwards, recruited participants through university bulletin boards (both online and offline) and job websites. Potential participants completed the Hospital Anxiety and Depression Scale (HADS) to measure their anxiety and depression levels. Our final sample included 119 young adults aged 20–39 years (average age = 27.04 years, SD = 3.49; 29.41% male). All participants showed borderline abnormal levels of anxiety and/or depression (HADS scores > 8).

3.3.3 Data collection. Participants were instructed to use the app consistently every day for eight weeks. Assessments were conducted biweekly through self-reported online surveys. We assessed participants using multiple validated scales: Hospital Anxiety and Depression Scale (HADS) [66], Beck Anxiety/Depression Inventories (BAI/BDI) [7, 43], Action and Acceptance Questionnaire (AAQ)

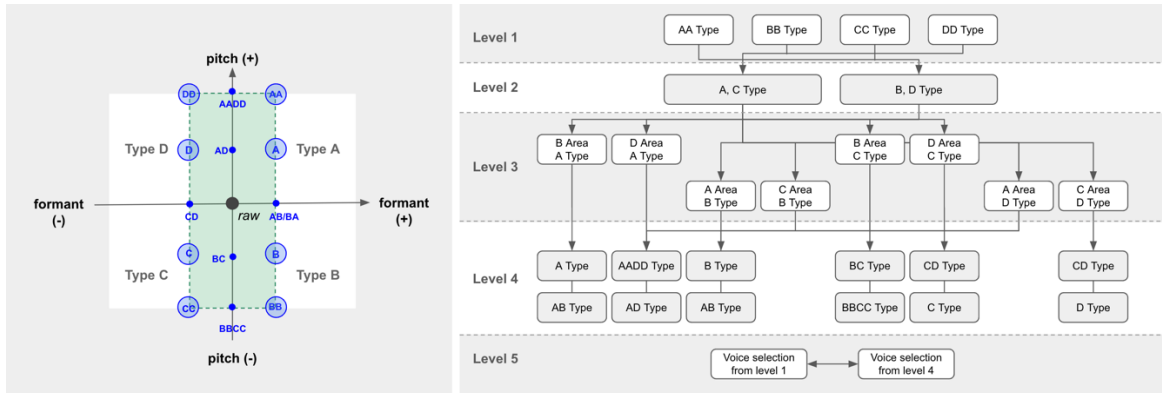


Figure 2: The algorithm consists of five stages designed to facilitate pairwise comparisons of 14 voice types with adjusted timbres using the Analytical Hierarchy Process (AHP). In level 1, participants are presented with four extreme voice types with distinct timbre adjustments. In level 2, they select three descriptive adjectives that reflect their preferred voice qualities, which are then used to determine the directional attributes of the timbre characteristics associated with their initial choice. Level 3 and 4 involves a more detailed comparison of voices with nuanced timbre adjustments, enabling participants to refine their preferences. Finally, in level 5, participants compare the extreme voice selected in the first level with the more detailed voice adjustments from the fourth level to identify their ideal voice type, which is perceived as the most comfortable and optimal.

[8], Connor-Davidson Resilience Scale (CD-RISC) [12], and Comparative/Internalized Self-criticism (CSC/ISC) [58]. These measures evaluated three key questions: the impact of self-talk alone on resilience and self-criticism, the effects of emotion-based content without vocalization, and potential interaction effects between self-talk and emotion-based content.

4 Findings

In this section, we present findings from both studies, including statistical and qualitative analyses. We report the results in chronological order of the studies.

4.1 Study 1: Self-talk as an Effective MoA for Alleviating Anxiety

4.1.1 Quantitative result. For each assessment, responses to the Likert-scale questions were compiled and analyzed by calculating the total score across items, ensuring a reliable measure of participants' experiences and changes over time. The Beck Anxiety Inventory (BAI) showed good reliability with a Cronbach's α of 0.844. We also used the Beck Depression Inventory (BDI) to measure depression and the Action and Acceptance Questionnaire (AAQ) to assess psychological flexibility. These measures also demonstrated good reliability (Cronbach's α : BDI = 0.834, AAQ = 0.831). Participants in the experimental group used the self-talk intervention an average of 4.40 days per week (62.83%) throughout the study. Using repeated measures ANOVA, we analyzed how self-talk affected participants based on their emotional state. Results showed that three weeks of self-talk intervention, which used Acceptance and Commitment Therapy (ACT) principles, significantly improved both anxiety ($F(1,55) = 6.402, p = .014$) and psychological flexibility ($F(1,55) = 6.539, p = .013$). Though both groups showed lower depression scores, the difference between groups did not reach statistical significance ($F(1,55) = 3.015, p = .088$).

4.1.2 Qualitative result: Self-talk, a refreshing experiencing, yet challenging to overcome the unfamiliarity. We conducted semi-structured interviews to understand participants' experiences with the three-week intervention. This interview format let us adapt our questions based on participants' responses [5, 43], which helped us explore their perspectives on previously unstudied aspects. The interviews covered five main areas: self-talk experience, self-talk content, chatbot-mediated delivery, overall satisfaction, and potential improvements.

Participants found the self-talk approach novel, with distinct experiences during speaking and listening phases. The intervention required users to speak content aloud before listening to it again. Several users noted privacy concerns in shared spaces: "I was fine with it in general, but when it comes to recording the voice, I could only do it at home. So this wasn't easy for me to use the app during my spare time" [P17], "I live in a dormitory, so I hardly did it" [P13]. Others viewed it positively, comparing it to reading aloud and noting its benefits: "Speaking slowly and thoughtfully becomes a refreshing experience for positive emotions" [P28].

The listening experience varied significantly. Those familiar with their recorded voice reported minimal discomfort: "I'm involved in acting and recording podcasts... Hearing my voice was enjoyable. The most significant aspect is the comfort and the feeling of solace" [P4]. Some found it therapeutically valuable: "When it played back what I had just said, sometimes I felt like I had a conversation with myself... It felt like I was comforting myself" [P7]. However, those unfamiliar with hearing their voice reported feelings of awkwardness: "In the beginning, I did listen a few times, but listening to my own voice was somewhat embarrassing... It was a bit uncomfortable hearing words I had spoken with my own lips" [P21]. Background music helped reduce this discomfort: "When the recording was played back with music at the end, it was quite fitting... I liked it because having music made it a bit more emotional" [P31].

Table 1: Study 2 result. Interaction effect of self-talk, emotion, and time on anxiety and resilience

DV	IV	F Statistics	Significance Level	Partial Eta-Squared
Anxiety (BAI)	Self-talk	$F(3,90) = 10.135$	$p = .002^{**}$	$\eta^2 p = .101$
	Emotion	$F(3,90) = 0.168$	$p = .683$	$\eta^2 p = .002$
	Time*Self-talk	$F(3,90) = 4.092$	$p = .008^{**}$	$\eta^2 p = .043$
	Self-talk*Emotion	$(3,90) = 5.548$	$p = .021^*$	$\eta^2 p = .058$
Resilience (CD-RISC)	Self-talk	$F(3,90) = 0.880$	$p = .351$	$\eta^2 p = .010$
	Emotion	$F(3,90) = 0.228$	$p = .634$	$\eta^2 p = .003$
	Time*Emotion	$F(3,90) = 4.890$	$p = .030^*$	$\eta^2 p = .052$
	Self-talk*Emotion	$F(3,90) = 4.862$	$p = .030^*$	$\eta^2 p = .051$
Comparative self-criticism (CSC)	Self-talk	$F(3,90) = 1.172$	$p = .282$	$\eta^2 p = .013$
	Emotion	$F(3,90) = 0.791$	$p = .376$	$\eta^2 p = .009$
	Time*Emotion	$(3,90) = 1.869$	$p = .175$	$\eta^2 p = .020$
	Self-talk*Emotion	$F(3,90) = 11.842$	$p < .001^{***}$	$\eta^2 p = .116$

^a Note: $^*p < .05$, $^{**}p < .01$, $^{***}p < .001$. IV: Independent Variable; DV: Dependent Variable; BAI=Beck Anxiety Inventory; CD-RISC=Connor-Davidson Resilience Scale

All in all, self-talk emerged as distinct from other practices like mindfulness meditation or therapeutic conversations. Participants valued it particularly as a tool for addressing negative thoughts as they arose, though experiences varied based on users' comfort with hearing their recorded voice.

4.1.3 Design implication. Users' experiences with self-talk showed mixed reactions rather than clear positive or negative patterns. However, the three-week intervention generally led to decreased anxiety and improved psychological flexibility. Based on these findings, we identified key design implications for our main study. Our first design implication focuses on addressing voice-related discomfort. We need to develop methods that reduce users' unease when listening to their recorded voice, particularly for those unfamiliar with hearing themselves speak. Our second design implication concerns improving the delivery method. The intervention's format itself can create discomfort during long-term use, especially when usage patterns become obvious to users. The delivery method needs refinement to support sustained engagement.

4.2 Study 2: Enhancement of One's Own Voice Experience While Alleviating Anxiety

4.2.1 Quantitative result. We conducted a study with two factors (Self-talk vs. Non-self-talk $\times$ Emotion-based vs. Non-emotion-based) to examine how self-talk affects anxiety when combined with emotion-based content, and to identify which elements contribute to anxiety reduction. From our initial 119 participants, we analyzed data from 94 individuals, as some participants did not complete the longitudinal study or missed survey responses.

Participants in each condition engaged with the intervention at varying frequencies. The Self-talk $\times$ Emotion-based group used the intervention an average of 4.17 days per week (59.50%), while the Emotion-based only group had a slightly higher usage at 4.31 days per week (61.56%). The Self-talk only group engaged with the intervention 3.92 days per week (56.06%), whereas the Non-self-talk $\times$ Non-emotion-based group had the average usage at 4.60 days per week (65.66%). We used repeated measures ANOVA to analyze

how self-talk interacts with emotion-based content over time. We measured anxiety at five points: week 0, week 2, week 4, week 6, week 8 (post-intervention) as the within-participant variable. The between-participants variable included four conditions: self-talk, non-self-talk, emotion-based, and non-emotion-based. The analysis revealed two significant effects. Self-talk alone showed a significant effect on anxiety reduction ($F(3,90) = 10.135$, $p = .002$). We also found a significant interaction effect between self-talk and emotion-based content on anxiety ($F(3,90) = 5.548$, $p = .021$) (Table 1).

We conducted another repeated measures ANOVA to examine how self-talk and emotion-based content affected resilience (CD-RISC) and comparative self-criticism (CSC). For resilience, emotion-based content showed a significant effect ($F(3,90) = 4.890$, $p = .030$), and we found a significant interaction effect when combined with self-talk ($F(3,90) = 4.862$, $p = .030$). For comparative self-criticism, neither self-talk nor emotion-based content alone produced significant changes. However, the combination of these two factors showed a significant interaction effect ($F(3,90) = 11.842$, $p < .001$).

We found no significant effects of self-talk and emotion-based content on depression (BDI), psychological flexibility (AAQ), or internalized self-criticism (ISC). We then used simple linear regression to examine how resilience affects anxiety. The regression model proved significant ($F = 28.091$, $p < .001$) and explained 23.2% of the variance ($R^2 = .232$). The Durbin-Watson value of 2.164 confirmed that the residuals were independent. Results showed that resilience had a significant negative relationship with anxiety ($\beta = -0.198$, $p < .001$), indicating that higher resilience levels correspond with lower anxiety levels (Table 2).

4.2.2 Qualitative result: Facilitating natural self-talk through voice equalization. Our findings from Study 1 confirmed that self-talk effectively reduces anxiety. Participants noted that using their own voice for self-talk led to deeper self-acceptance and emotional understanding compared to external support. By speaking and listening to content in their own voice, users engaged with the messages more deeply than they would through reading alone.

Table 2: : Study 2 result. Linear regression analysis between anxiety(BAI) and resilience(CD-RISC)

DV	IV	B	S.E.	β	t	p
Anxiety (BAI)	(Constant)	-3.553	.595	N/A	-5.973	< .001***
	CD-RISC	-0.198	.037	-.482	-5.300	< .001***
$F = 28.091$ ($p < .001$), $R_2 = .232$, D-W = 2.164						

^a Note: * $p < .05$, ** $p < .01$, *** $p < .001$. IV: Independent Variable; DV: Dependent Variable; B=Unstandardized B; S.E=Standard error; β =Standardized coefficients beta; BAI=Beck Anxiety Inventory; CD-RISC=Connor-Davidson Resilience Scale

Participants valued the extended reflection time: *"By listening and re-listening, the time spent thinking about the message seemed to extend, which was beneficial"* [P19]. The experience created a unique form of self-talk, with one user noting it felt like *"replaying something to myself"* [P20]. Unlike traditional meditation apps, this approach fostered self-discovery, as one participant observed: *"getting to know myself while listening to my own voice"* [P13].

Voice modification proved crucial in reducing psychological barriers to self-talk. Users found ways to overcome their initial discomfort with hearing their recorded voice. As one participant explained: *"I used to find my voice strange when listening to it, but now I can choose from various voice options"* [P20]. Another noted that *"the modulated voice is just a modulated voice, so it doesn't matter"* [P8], suggesting that voice modification created a safer environment for self-talk.

The voice modification feature also enhanced user engagement. Some participants who initially avoided their recorded voice found enjoyment through this feature: *"At first, I found listening to my voice too awkward and unpleasant, but as I tried the voice conversion feature and experimented with various options, I found it quite entertaining"* [P13]. Though some viewed voice modification as merely novel, it effectively reduced resistance to self-talk. Starting with *"I initially tried it out of curiosity"* [P21], users gradually became more engaged. Some even developed greater acceptance of their natural voice: *"Initially, I did not like my voice and thought a voice actor would be better, but after recording, I realized my voice is somewhat bearable"* [P22].

5 Discussion

5.1 Interpretation of the Result

5.1.1 The importance of methods to enhance engagement in self-talk. Our pre-study interviews revealed that users' experiences with self-talk centered on hearing their recorded voice. People who do not regularly work with voice recordings often find this experience unsettling. This feeling of discomfort occurs because we normally hear our voice through both air and bone conduction, which creates a noticeable difference when listening to only the recorded version [37]. To address this issue in the main study, we used the Analytic Hierarchy Process (AHP) to help users find their preferred voice filter, ensuring self-talk more comfortable. The approach proved successful: 83% of participants in the self-talk groups used their ideal voice filter, whilst only 4% (two participants) did not use the filter at all. These findings taught us that rather than imposing self-talk, we should encourage its natural adoption to maintain user engagement. Whilst self-talk can effectively reduce anxiety, increase resilience, and decrease comparative self-criticism

– particularly when it matches the user's situation – these benefits diminish if users cannot maintain the practice.

5.1.2 The relationship between self-talk, resilience, and anxiety. This study aimed to reduce anxiety through self-talk as a novel MoA. We incorporated emotional data collection and personalized ACT content to enhance engagement. Both studies demonstrated consistent anxiety reduction effects. Anxiety, often stemming from future-oriented threat perceptions [24], can be mitigated through repetitive self-talk by shifting focus to the present moment [11]. Our main study revealed that while self-talk alone didn't improve resilience, combining it with emotion-based ACT content showed positive impacts and synergistic effects. Content matching current emotional states proved more effective than unrelated self-talk, particularly in promoting objective self-reflection. This approach could potentially enhance self-recovery through repeated training. While extended self-talk didn't increase psychological flexibility, improvements were observed in resilience and reduced comparative self-criticism. This aligns with previous research showing self-talk's positive impact on psychological resilience through self-esteem and self-compassion mediation [27, 52].

5.2 Implications for HCI Research

Our findings advance several implications for HCI research in digital mental health interventions:

- **Voice-Based Therapeutic Interaction.** While traditional voice interfaces prioritize functional aspects, our findings demonstrate the critical role of emotional sensitivity in therapeutic voice interactions. Studies show that voice modification can significantly impact user engagement in healthcare applications. Further exploring this voice-engagement relationship, research in human-computer interaction suggests that self-voice may function as an effective mechanism for encouraging goal-directed behaviors [36]. Our voice equalization approach extends this understanding by introducing a framework for psychologically comfortable voice interactions in therapeutic contexts.
- **Behavioral Nudges in Digital Therapeutics.** Our implementation of self-talk as a behavioral nudge builds on digital nudging research while introducing novel applications in therapeutic contexts. Previous work has shown that subtle digital interventions can effectively modify health behaviors [50]. Our research extends this by demonstrating how voice-based nudges can be naturally integrated into therapeutic frameworks without compromising user autonomy.
- **Adaptive Personalization in Mental Health Applications.** The varying responses to voice modifications align

with findings on adaptive interfaces in healthcare. Prior HCI research has shown that self-referential voice experiences through voice manipulation can contribute to enhanced self-awareness and emotional regulation abilities [13, 68]. Our systematic approach to voice preference optimization contributes to the growing body of work on personalization in digital therapeutics [32]. The results suggest that real-time adaptation of interaction modalities is crucial for maintaining therapeutic engagement.

6 Conclusion

This research investigated self-talk as a novel therapeutic mechanism and behavioral nudge in digital anxiety interventions. Our findings demonstrate three key contributions: (1) self-talk's effectiveness as a Mechanism of Action for anxiety reduction, (2) its role as a behavioral nudge promoting therapeutic engagement, and (3) the importance of voice equalization in enhancing user interaction. Our studies show that self-talk functions effectively as both a therapeutic tool and subtle behavioral nudge, with our voice equalization algorithm addressing a critical adoption barrier.

While our findings demonstrate promise, several limitations warrant consideration. The study focused primarily on immediate anxiety reduction effects, leaving longitudinal impacts unexplored. Additionally, individual differences in voice preference and their impact on therapeutic outcomes require further investigation.

These findings have important implications for DTx design, demonstrating how integrating nudge principles with therapeutic mechanisms can enhance intervention effectiveness. Future research should explore long-term effectiveness of self-talk nudges and examine cultural and individual differences in voice preference to enhance personalization and strengthen therapeutic outcomes.

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